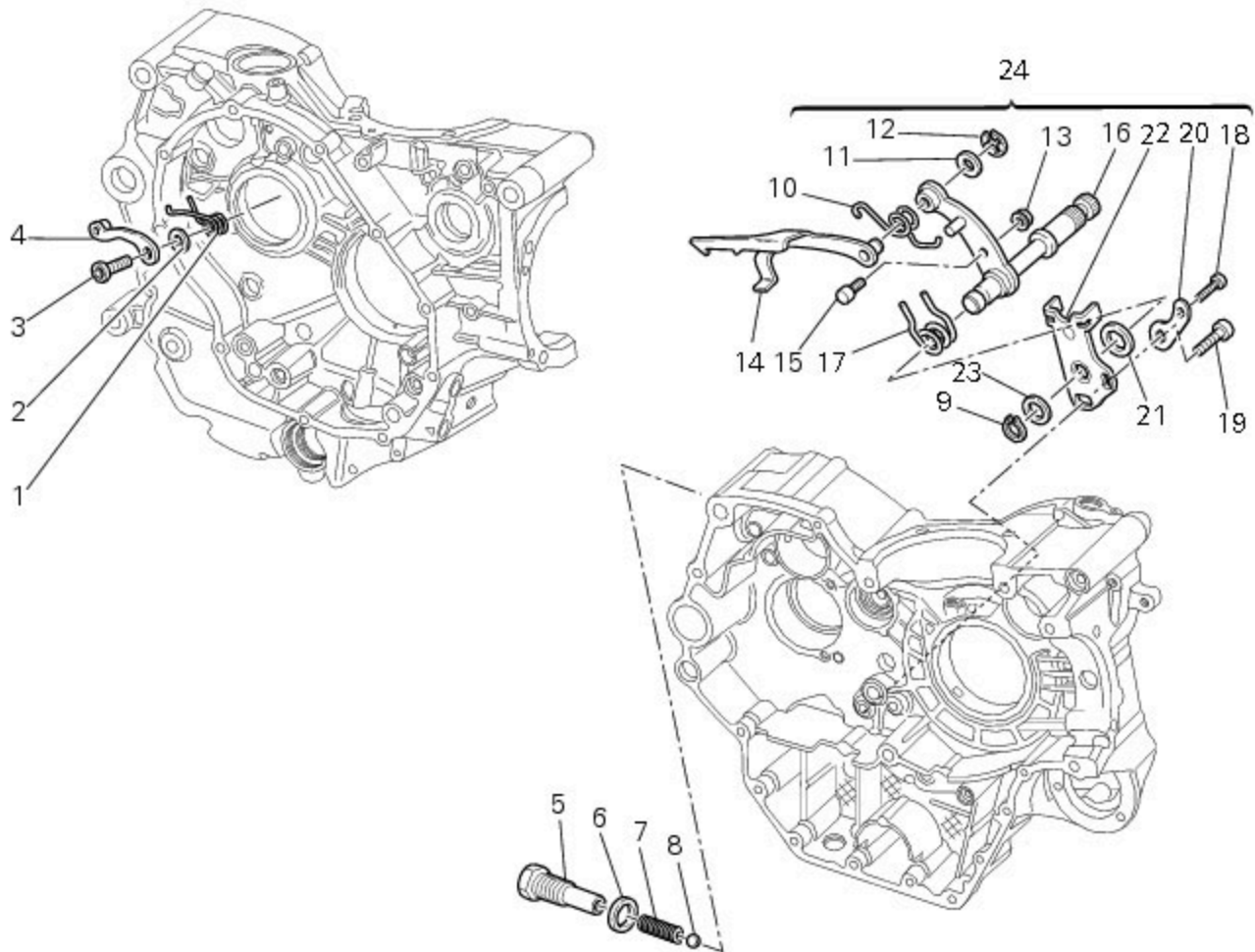


7.1 -Gearbox assembly: gearchange mechanism



- 1 Return spring
- 2 Washer
- 3 Banjo bolt
- 4 Gear pawl assembly
- 5 Interlock plunger holder
- 6 Sealing washer
- 7 Detent ball spring
- 8 Ball
- 9 Circlip
- 10 Selector claw return spring
- 11 Shim washer
- 12 Ring
- 13 Nut
- 14 Selector drum control fork
- 15 Gearchange lever pin
- 16 Shaft with gearchange lever arm
- 17 Gearchange lever arm return spring
- 18 Screw
- 19 Screw
- 20 Spacer
- 21 Locating ring
- 22 Stop plate
- 23 Shim washer
- 24 Gearchange mechanism

1100 ABS [GEARCHANGE CONTROL](#)

1100 S ABS [GEARCHANGE CONTROL](#)

Important

Bold reference numbers in this section identify parts not shown in the figures alongside the text, but which can be found in the exploded view diagram.

Removal of the gear selector lever

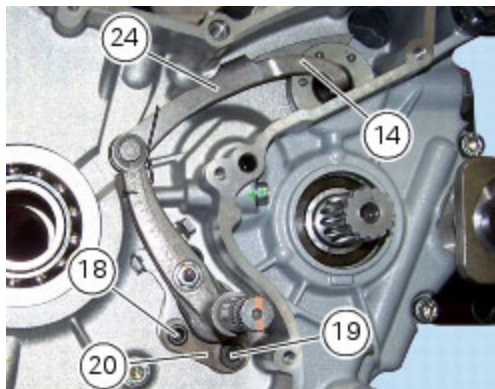
Operation	Section reference
Drain the engine oil	D 4, Changing the engine oil and filter cartridge
Remove the gearchange control	F 5, Removal of the gearchange control
Remove the clutch slave cylinder	F 2, Removal of the clutch slave cylinder
Remove the front sprocket cover	G 8, Removing of the front sprocket
Remove the generator cover and flywheel-generator assembly	N 8, Removal of the generator cover

Undo the gear selector mechanism (24) retaining screws (18) and (19).

Remove screws (18) and (19), spacer (20) and withdraw the gear selector mechanism complete with the gear lever shaft, spring and plate.

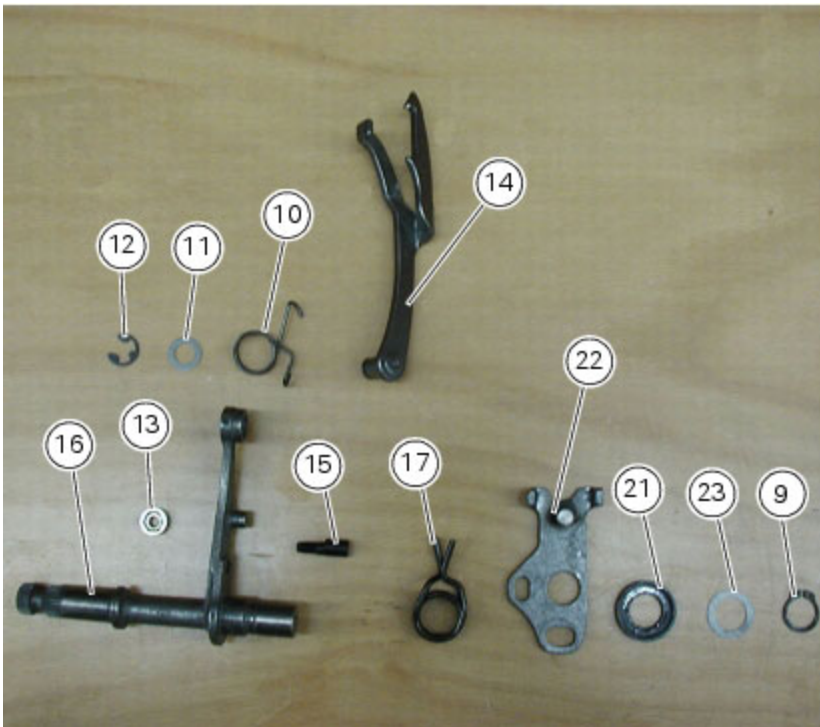
Important

Visually inspect the gear selector claw (14) for wear, particularly around the area where it contacts the selector drum.



Disassembly of the gear selector lever

If it is necessary to renew any components, disassemble the gearchange mechanism as shown in the exploded view.

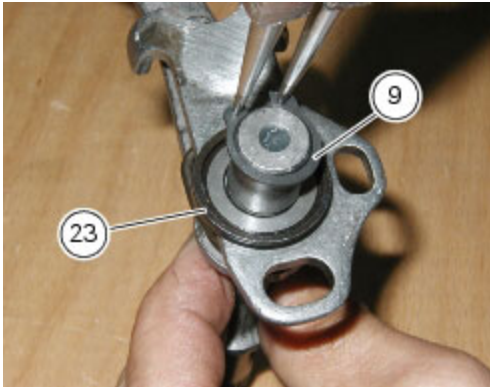
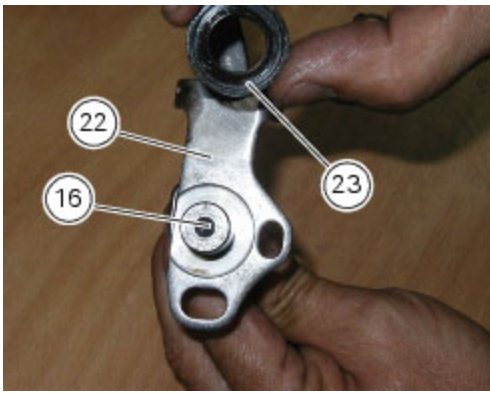


Reassembly of the gear selector lever

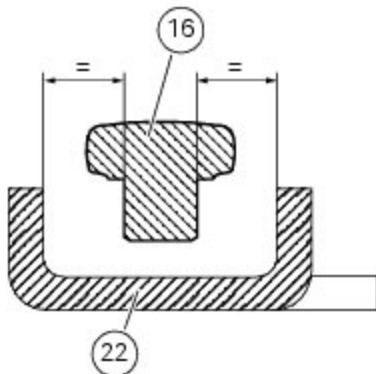
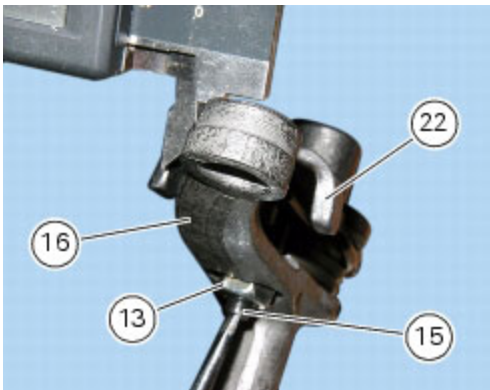
If the components of the gearchange mechanism were dismantled on removal, fit the spring (17) on the shaft (16) in the position indicated in the photo. Grease the threaded end of the eccentric gearchange lever pin (15) and install it on the upper part of the gearchange lever, securing it with the nut (13). Screw the nut on by hand (do not tighten)).



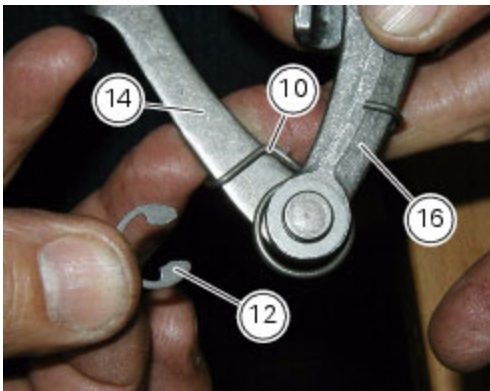
Insert the shaft (16) in the plate (22), fit washer (23) and secure with circlip (9).



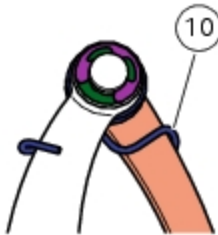
Ensure that the shaft with lever (16) is positioned correctly relative to stop plate (22). The edges of the plate must be equidistant from the pin of the lever; to adjust the position, loosen the lock nut (13) and turn eccentric pin (15).



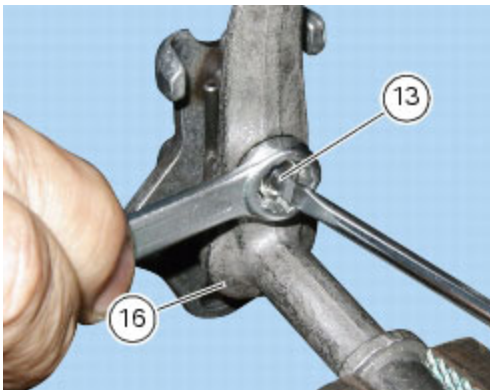
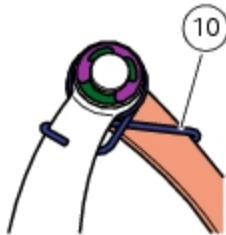
Position the spring (10) between the shaft (16) and the fork (14).
 Fit the ring (12) to secure the assembly.
 Check that the spring (10) is installed correctly as shown in the figure.
 Tighten the nut (13) to the specified torque (Sect. C 3, [Engine torque settings](#)).



Incorrect fitting



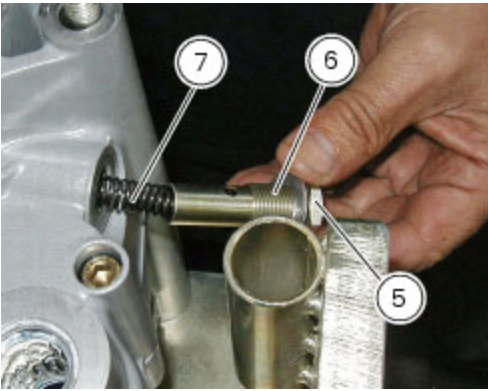
Correct fitting



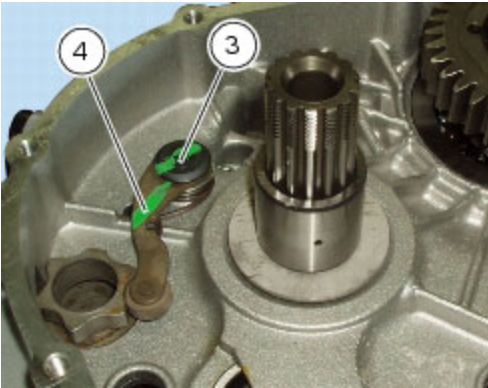
Disassembly of gear interlock plunger and pawl assembly

Operation	Section reference
Drain the engine oil	D 4, Changing the engine oil and filter cartridge
Disconnect the oil pressure sensor	P 5, Checking the front and rear brake light switches, neutral light switch, oil pressure switch and clutch switch
Remove the clutch assembly	N 6.1, Removal of the clutch
Remove the clutch cover	N 6.2, Removal of the clutch cover
Remove the oil pump	N 2.1, Removal of the oil pump
Remove the primary gear	N 6.3, Removal of the primary drive gear

Unscrew the interlock plunger screw (5) and remove the seal (6), spring (7) and the detent ball (**8**).



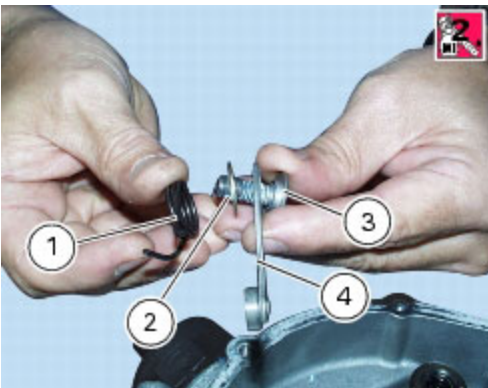
Unscrew the clutch-side crankcase half screw (3) and remove the pawl (4), washer (2) and spring (1).

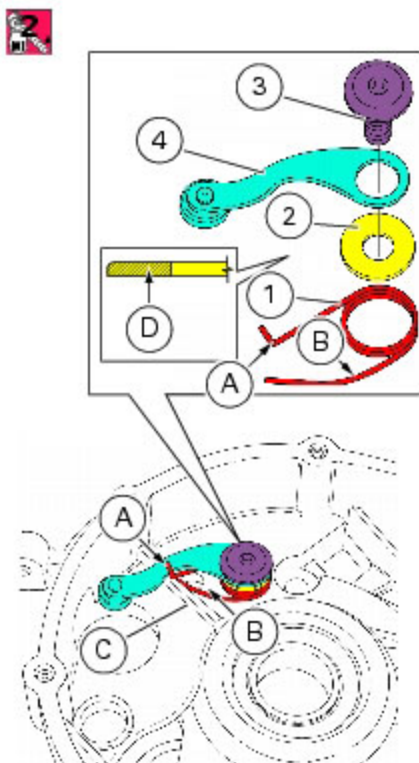


Refitting the gear interlock plunger and pawl assembly

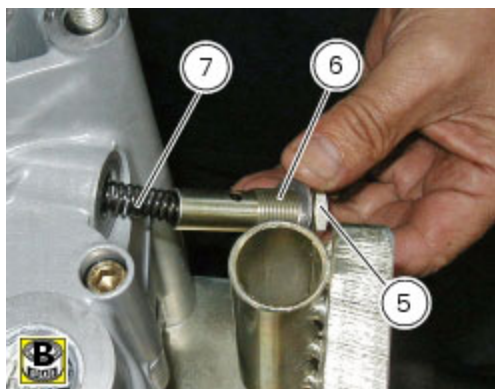
On the fastening screw (3), fit the gear pawl lever (4), orienting it as shown in the figure, the washer (2) with the square edge side (D) facing the clutch-side crankcase half, and the spring (1), positioning it so that the hook end (A) is facing the gear pawl lever. Locate the hook end (A) of the spring on the gear pawl lever as shown in the figure. Apply threadlocker to the screw thread.

Start the screw in the crankcase half. Position end (B) of the spring so that it rests against rib (C) of the crankcase half, as shown in the figure. Tighten the screw (3) to the specified torque (Sect. C 3, [Engine torque settings](#)). Check that the spring functions correctly by manually moving the gear pawl lever.





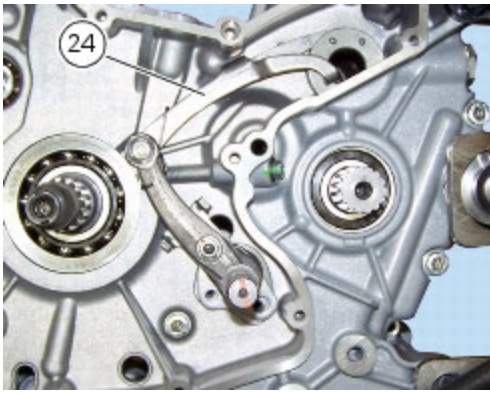
Apply grease and then fit the ball (8), spring (7), and seal (6) into interlock plunger (5) and screw the interlock plunger assembly into the crankcase. Tighten the gear interlock plunger to the specified tightening torque (Sect. C 3, [Engine torque settings](#)) and screw the gear interlock plunger assembly on the crankcase.



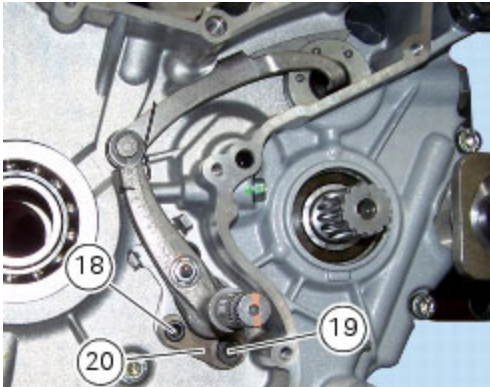
Operation	Section reference
Refit the primary gear	N 6.3, Fitting the primary drive gear pair and checking backlash
Refit the clutch cover	N 6.2, Reassembly of the clutch cover
Refit the complete clutch assembly	N 6.1, Reassembling the clutch
Connect the oil pressure sensor	P 5, Checking the front and rear brake light switches, neutral light switch, oil pressure switch and clutch switch
Refill the engine with oil	D 4, Changing the engine oil and filter cartridge

Refitting the gear selector lever

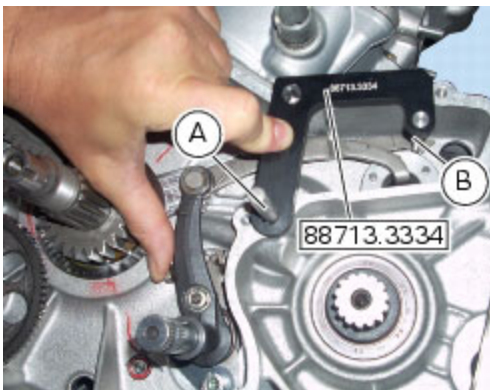
Position the gearbox drum selector fork in the centre of the gear rollers.
Position the whole gear selector lever (24) into the chain-side crankcase half.



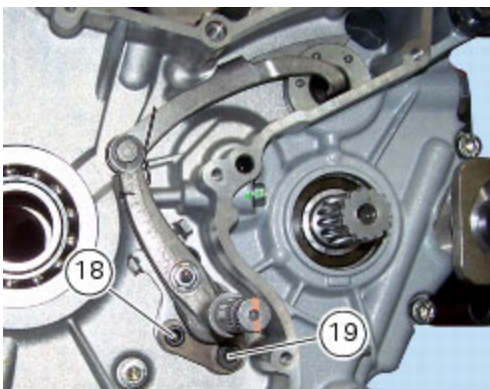
Insert the screws (18) and (19) with the spacer (20).
Temporarily fit gear change lever (or a service lever) and sprocket and shift to neutral gear.



Place the tool **88713.3334** on the gear claw.
Place the tool inserting the pushrod (A) into the tool hole, block the pin (B) of the tool in the gear claw pressing with the hand in the point (C) towards the right.



Tighten the gear selector lever fixing screw (19) and then the (18) (Sect. C 3, [Engine torque settings](#)).

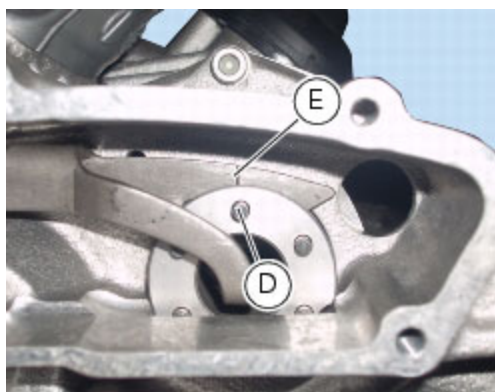


Remove the tool and check that the pin (D) placed on the gear drum is axially to the notch (E) on the gear claw.

With the gearbox in neutral, check that the lever travel is the same when shifting up and down. The same should apply when a gear is engaged.

Operate the gearchange lever and turn the front sprocket at the same time to check that all the gears engage when shifting up and down.

Remove the gear change lever.



Operation	Section reference
Refit the flywheel/generator assembly and generator cover	N 8, Refitting the flywheel-alternator assembly
Refit the front sprocket cover	G 8, Refitting the front sprocket
Refit the clutch slave cylinder	F 2, Refitting the clutch slave cylinder
Refit the gearchange control	F 5, Refitting the gearchange mechanism
Fill the engine with oil	D 4, Changing the engine oil and filter cartridge